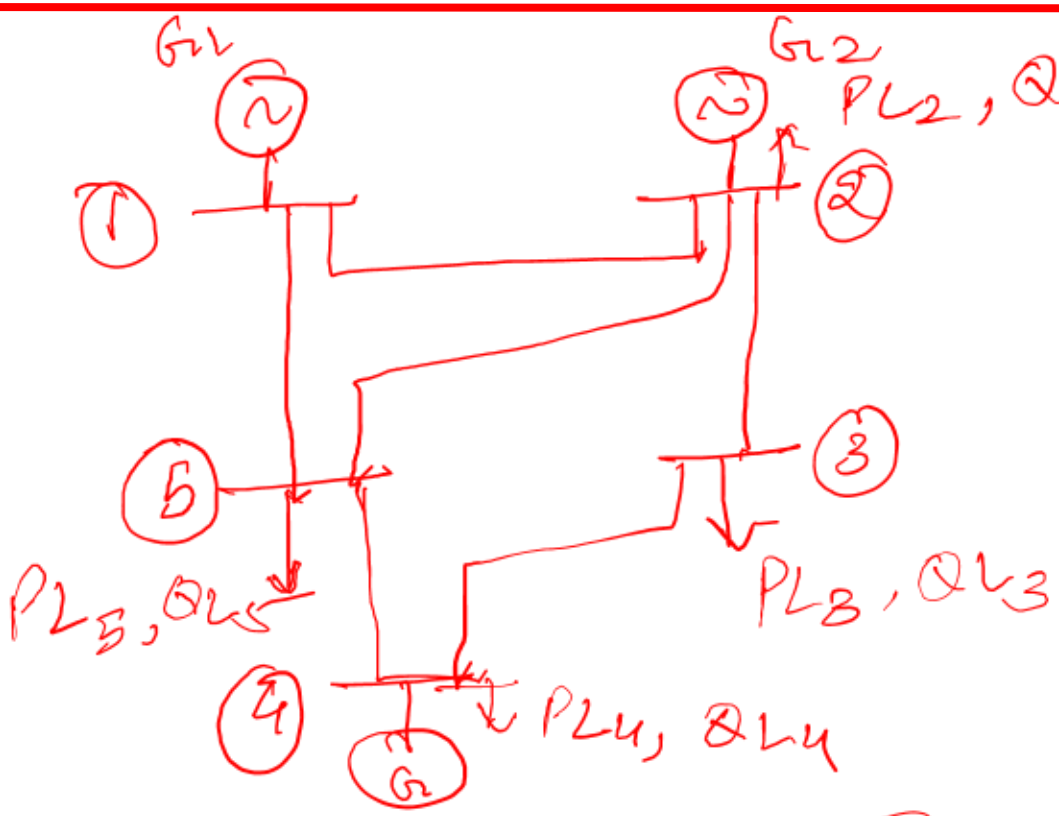
AENL 226
Power Electronics and Power Systems

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1, 2, 4 ⇒ Gen Bus/Load

3, 5, Load Bus

G2 ← Gen-No (2)

⇒ 5-Bus Power System

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$$P_{G1} + P_{G2} + P_{G4} = P_{L2} + P_{L3} + P_{L4} + P_{L5}$$
$$+ P_{loss\,1-2} + P_{loss\,2-3} + P_{loss\,3-4}$$
$$+ P_{loss\,4-5} + P_{loss\,5-1} + P_{loss\,2-5}$$

Representation of the various components such as generator (synchronous)



voltage source in series with End ⁿ reactance

Per-unit values put every component on one scale

- **The per-unit system expresses electrical quantities as fractions of selected base values.**

$$\text{Value in p.u.} = \frac{\text{Actual value}}{\text{Selected base value}}$$

A per-unit quantity is dimensionless. 1.0 p.u. means "equal to base"; 8% means 0.08 p.u.

- Equipment data become directly comparable
- Transformer ratios disappear from the impedance network
- Most operating values remain close to 1.0, so errors stand out

Source: Grainger & Stevenson, 2017.

Changing the Base of Per-Unit Quantities

- Components may have per-unit impedances expressed on different bases.
- All impedances in the same part of a power system must use a common base.
- Base conversion is therefore necessary for correct system calculations.

$$\text{Per-unit impedance} = \frac{(\text{actual impedance, } \Omega) \times (\text{base kVA})}{(\text{base voltage, kV})^2 \times 1000}$$

- Per-unit impedance is directly proportional to base kVA.
- It is inversely proportional to the square of base voltage.
- Changing either base kVA or base kV changes the per-unit impedance.

$$\text{Per-unit } Z_{\text{new}} = \text{Per-unit } Z_{\text{given}} \left(\frac{\text{base kV}_{\text{given}}}{\text{base kV}_{\text{new}}}\right)^2 \left(\frac{\text{base kVA}_{\text{new}}}{\text{base kVA}_{\text{given}}}\right)$$

- Start with the given per-unit impedance.
- Multiply by the square of the voltage base ratio.
- Multiply by the apparent power base ratio.
- The result is the impedance on the new base.

Choose Sbase and Vbase; the other bases follow

For a balanced 3-phase system, use three-phase apparent power and line-to-line voltage.

Select independently

Sbase : common MVA base

VLL,base : voltage base at one bus

Propagate Vbase through transformers using their rated voltage ratios.

Derived quantities

$$I_{\text{base}} = \frac{S_{\text{base}}}{\sqrt{3}\, V_{\text{LL,base}}}.$$

$$Z_{\text{base}} = \frac{V_{\text{LL,base}}^2}{S_{\text{base}}}.$$

$$\boxed{Z_{\text{base}}\,[\Omega] = \frac{(V_{\text{base}}\,[\text{kV}])^2}{S_{\text{base}}\,[\text{MVA}]}}$$

Source: Grainger & Stevenson, 2017.



With consistent bases, an ideal transformer disappears

Choose voltage bases in the same ratio as the transformer's rated voltages:

$$V_{base,H} / V_{base,L} = N_H / N_L$$

115 kV → **13.8 kV**

What stays the same

The transformer's p.u. impedance is identical when referred to either side.

What simplifies

The ideal turns ratio no longer appears in the p.u. impedance diagram.

Result: one continuous network across several voltage levels.

Convert every impedance to the chosen system base

$$Z_{\text{pu,new}} = Z_{\text{pu,old}} \left(\frac{S_{\text{base,new}}}{S_{\text{base,old}}} \right) \left(\frac{V_{\text{base,old}}}{V_{\text{base,new}}} \right)^2$$

Worked example

The reactance of a generator is X ″ = 0.25 per unit on its nameplate base of 18 kV and 500 MVA.The new base is 20 kV and 100 MVA. Find X ″ on the new base.

Generator reactance: X″ = 0.25 p.u. on 500 MVA, 18 kV
Required system base: 100 MVA, 20 kV

$$X''_{\text{new}} = 0.25 \left(\frac{100}{500} \right) \left(\frac{18}{20} \right)^2$$

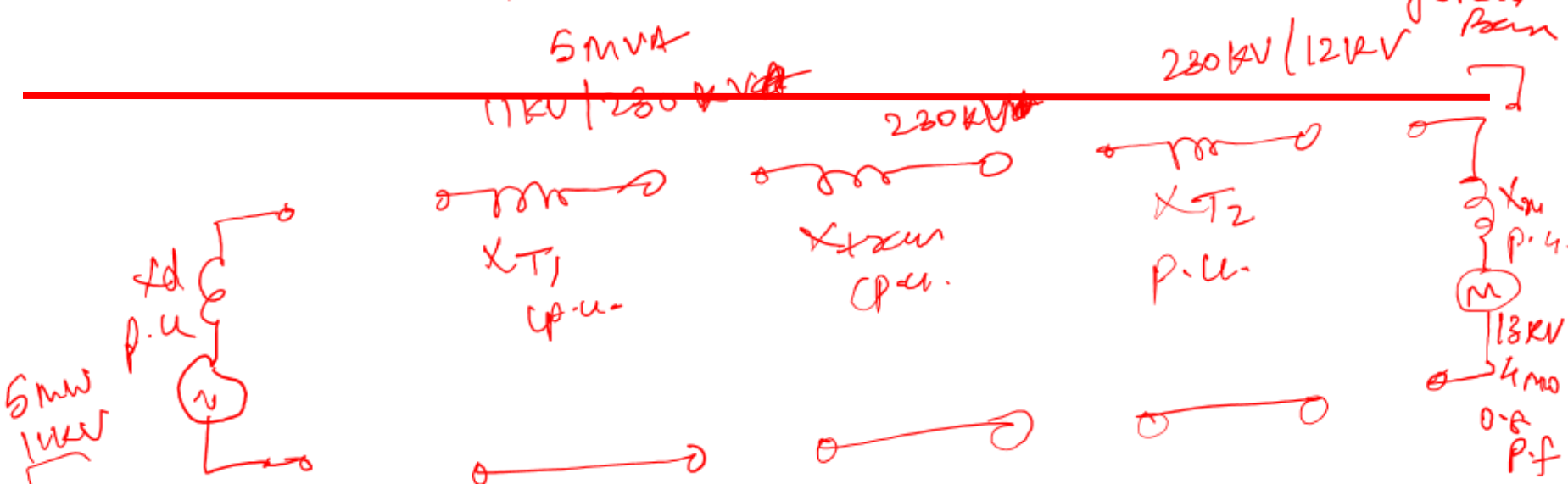
$$= 0.0405 \text{ p.u.}$$

Check the direction: a larger voltage base reduces p.u. impedance; a smaller MVA base also reduces it.

Source: Grainger & Stevenson, 2017.



ONE ← of the rated/base to be selected as overall system Base



p.u. values of individual devices are in different basis ←

A → Individual equivalent circuits can be connected together only if all parameters are brought to same base

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(B) The values of various equivalent circuit parameters are given at rating of that particular device (Data sheet or Testing)

Select a Base MVA or KVA and voltage ?

One of the rating of the existing devices